

Genesis: Physically Grounded Multi-Agent Artificial Life Simulation with Emergent Social and Cognitive Dynamics

Vinay Khosya¹

¹Independent Researcher , vinayroyale123@gmail.com
genesis.vinaykhosya.com , ORCID: 0009-0002-5232-5432

July 2026

Engine v1.5.0 | Schema phase8.4

Preprint Manuscript — arXiv cs.NE / cs.MA / q-bio.PE

Abstract

Artificial life (ALife) simulations provide fundamental insights into emergent evolutionary biology, social coordination, and cognitive adaptation under environmental stressors. However, traditional computational substrates treat the physical landscape as a uniform setting rather than an active experimental variable, decoupling evolutionary dynamics from spatial geography. We present **Genesis**, an agent-based artificial life simulator built from first principles to evaluate how physically derived environments—incorporating multi-octave Perlin elevation noise, latitude-driven thermodynamics, wind-advection precipitation, Priority-Flood hydrology, and a Whittaker biome matrix—shape evolutionary selection and cognitive architecture. Sixteen founding agents divided across four geographically distinct colonies are initialized on a 1024×1024 grid with identical founder genome distributions ($\text{avg_genetic_diversity} = 0.0$) and observed over longitudinal simulation runs spanning up to 326,075 ticks (905.8 simulated years). Agents inherit 14 continuous genomic traits governing physiological decay, perception, and neural plasticity, operating via a multi-drive homeostatic deliberation system with spatial predictor networks. No behavioral outcomes or social structures are scripted. In a pilot experiment under extreme resource scarcity ($S=3.0$, $n=1$ run, seed 1720), we demonstrate: (1) **Geographic Selection**: initial biome placement determined colony survival, with the riparian colony accumulating 55.5×10^6 water units and persisting for 905.8 years, while arid and boreal colonies underwent complete extinction; (2) **Cognitive Consolidation**: directional selection collapsed exploratory novelty-seeking ($g_{\text{novelty_seeking}}: -72.4\%$) and neural learning rate ($g_{\text{learning_rate}}: -95.6\%$) as agents transitioned from active spatial learning to automated procedure execution; (3) **Technological Buffering**: construction of 3,009 shelters rendered thermoregulation ($g_{\text{thermoregulation}}: -62.4\%$) evolutionarily neutral without survival penalty. These findings establish that environmental heterogeneity is a primary engine of selection, cognitive specialization, and cultural buffering in multi-agent artificial life systems. **All results represent preliminary single-run observations.** Complete source code, simulation schemas, raw datasets, and interactive replays are openly accessible at <https://genesis.vinaykhosya.com>.

Contents

1	Introduction	4
1.1	The Problem of Geography in Artificial Life	4
1.2	Why Genesis Exists: Motivation	4
1.3	Research Questions	4
1.4	Contributions	5
1.5	Reproducibility Statement	5
1.6	Scope and Limitations	5
2	Related Work	5
2.1	Avida	5
2.2	Sugarscape	6
2.3	NEAT and Neural Evolution Platforms	6
2.4	OpenAI Five and Multi-Agent RL	6
2.5	Geb and Open-Ended Evolution Systems	6
3	The Physical World Model	7
3.1	Terrain Generation	7
3.2	Temperature Model	7
3.3	Rainfall – Wind Moisture Advection	7
3.4	Biome Classification	7
3.5	Hydrology – Priority-Flood Drainage	8
3.6	Resource and Habitability Layers	8
4	Agent Architecture	8
4.1	Genome	8
4.2	Cognitive Architecture & Deliberation Loop	8
4.3	Drive Dynamics, Mortality, and Reproduction	8
4.4	Infrastructure	9
5	Experimental Protocol	9
5.1	Experimental Procedure	9
6	Pilot Study Results	10
6.1	Colony Setup & Survival Timeline	10
7	Results	11
7.1	Geographic Selection as a Viability Determinant	11
7.2	Directional Genomic Selection & Co-Evolutionary Interpretation	11
7.3	Cognitive Consolidation: The Learning Rate Collapse	12
7.4	Technological Buffering of Thermoregulation	13
7.5	Mortality Profile	13
8	Discussion	13
8.1	Geography as Independent Variable	13
8.2	The Scarcity–Curiosity Trade-Off	13
8.3	Cognitive Consolidation and Procedural Fixation	13

9	Open Questions	14
10	Code, Data, and Replay Availability	14
11	Limitations	14
12	Conclusion	15

1. Introduction

1.1 The Problem of Geography in Artificial Life

Every prior artificial life system made the same structural assumption: that the environment is uniform.

Avida places digital organisms in a flat computational lattice [10, 11]. Sugarscape distributes sugar on a fixed 50×50 grid [7]. Conway’s Game of Life operates on a homogeneous plane. Polyworld [16] and Tierra [12] simulate organism energetics in unmodeled topographies. Even modern multi-agent reinforcement learning platforms place agents in symmetric, procedurally balanced arenas [1].

This assumption is not merely a simplification—it forecloses an entire category of scientific investigation. In biological reality, geography is not noise. A mountain range creates a rain shadow. A rain shadow creates a desert. A desert means scarce fresh water. Scarce fresh water means selection pressure operates on thirst tolerance and spatial memory, not on general cognitive capacity.

The landscape is not a setting. It is an independent variable.

Genesis is an artificial life simulator designed from first principles around this insight. Before an agent is placed, the world is constructed: elevations emerge from multi-octave Perlin noise, temperatures from a latitude–elevation model, rainfall from wind moisture advection, rivers from Priority-Flood drainage, and biomes from the Whittaker temperature–rainfall matrix.

1.2 Why Genesis Exists: Motivation

Why build another artificial life simulator when established platforms such as Avida, Sugarscape, and Polyworld already exist? Because existing simulators structurally decouple cognition and evolution from the physical laws of the landscape. In Avida, organisms replicate on a flat grid; in Sugarscape, resources spawn on fixed schedules; in multi-agent RL arenas, environments are procedurally balanced. Genesis was built to test a hypothesis that cannot be evaluated in uniform substrates: that environmental heterogeneity—rain shadows, altitude thermoclines, river basins—is the primary engine of evolutionary selection and cognitive specialization.

1.3 Research Questions

1. **Viability:** Do geographically distinct initial colony placements produce systematically different survival outcomes when agent genomes are initialized with equal means?
2. **Evolutionary trajectory:** Does resource scarcity produce characteristic directional selection signatures across 14 heritable genomic traits?
3. **Technological buffering:** Does infrastructure construction decouple phenotypic fitness from the underlying genome, reducing selection pressure on previously critical traits?
4. **Cognitive consolidation:** Is there evidence of an evolutionary transition from high-plasticity active learning to low-plasticity procedural automation as ecological

niches become established?

1.4 Contributions

This work provides five primary contributions to artificial life and computational evolutionary biology:

1. **Physically Grounded ALife Platform:** A computational simulator where terrain, climate, hydrology, and biomes are derived from physical equations rather than hand-crafted rules, enabling geography to serve as an independent experimental variable.
2. **Genome-Driven Cognitive Architecture:** A multi-drive homeostatic agent model integrating 14 continuous heritable traits with a spatial predictor network and automated procedure formation.
3. **Long-Duration Reproducible Experiment Protocol:** A longitudinal telemetry framework tracking 300,000+ simulation ticks (900+ simulated years) capturing lineage extinction, technological shelter adoption, and genetic drift.
4. **Public Research Archive & Web Portal:** A searchable research portal (<https://genesis.vinaykhosya.com/archive>) hosting complete experiment datasets, JSON-LD metadata, and browser-based tick-by-tick replays.
5. **Open-Source Infrastructure:** Standardized schemas (`experiment_schema.json`) and Python execution pipelines for cross-platform simulation reproducibility.

1.5 Reproducibility Statement

To ensure exact computational reproducibility, every simulation run in Genesis adheres to strict execution standards:

- **Seed Determinism:** Procedural world generation and pseudo-random numbers are controlled by integer seed 1720.
- **Config Hash:** Full simulation parameters are logged in `config.json`.
- **Telemetry Archive:** Complete CSV logs and JSON event streams are mirrored in Supabase object storage.
- **License:** Source code is licensed under the MIT License; experimental datasets under CC BY 4.0.

1.6 Scope and Limitations

This paper reports one longitudinal pilot study ($n=1$, seed 1720, 326,075 ticks). All effect sizes and causal inferences are preliminary. Section 11 documents specific limitations.

2. Related Work

2.1 Avida

Avida [10, 11] demonstrated that self-replicating digital programs evolve increasing functional complexity under selective pressure, producing complex logical operations absent in

founding genomes. It established that Darwinian open-ended evolution is achievable in digital substrates.

However, Avida’s organisms occupy a uniform computational lattice with no spatial thermodynamics, climate, or hydrological resource gradients. Selection operates exclusively through instruction execution speed and replication rate, not physical survival against environmental hazards. This design prevents investigation of how environmental heterogeneity—such as rain shadows or thermoclines—creates spatially structured selection pressure. Genesis addresses this gap by making the physical world the primary experimental control.

2.2 Sugarscape

Sugarscape [7] introduced spatial heterogeneity into agent-based modelling by placing sugar and spice resource peaks on a 50×50 grid, enabling foundational studies of migration, wealth inequality, and trade emergence.

However, Sugarscape’s environment is static: resource replenishment follows fixed algebraic rules, climate is absent, and terrain elevation does not exist. Genesis extends this paradigm by synthesizing a physically derived world where resource distribution emerges naturally from terrain, wind moisture advection, and river hydrology rather than manual placement.

2.3 NEAT and Neural Evolution Platforms

NeuroEvolution of Augmenting Topologies [13] and novelty search algorithms [9] demonstrated that neural network topologies and behavioral diversity can be evolved alongside connection weights to discover specialized controllers for robotics, pole balancing, and game agents.

These platforms operate primarily as fitness-landscape optimization tools for fixed tasks rather than simulating open-ended multi-agent survival. Genesis incorporates genomic encoding of 14 cognitive and physiological traits, but embeds them within an unscripted survival ecology rather than an objective-function optimization loop.

2.4 OpenAI Five and Multi-Agent RL

Multi-agent reinforcement learning systems such as OpenAI Five [4], AlphaStar [14], and emergent autotricula platforms [1, 6] demonstrated emergent teamwork, strategic bluffing, and long-horizon coordination in complex multi-player games.

However, these systems operate in symmetric, human-engineered game arenas with pre-defined reward structures. Agent behaviors are trained via gradient descent—they do not evolve heritably across generations. Genesis differs in that all behavioral drives are encoded in a genome transmitted with mutation across generations, capturing long-timescale evolutionary selection that RL frameworks do not model.

2.5 Geb and Open-Ended Evolution Systems

Geb [5], Polyworld [16], and Tierra [12] pursued whether open-ended complexity growth occurs indefinitely in digital substrates.

While these systems demonstrated sustained evolutionary activity, they operated in minimal-physics environments. Genesis occupies a distinct scientific niche: a geographically grounded study of survival, social clustering, and cognitive evolution under physically realistic selection pressures.

3. The Physical World Model

Genesis constructs a physically derived world before any agent is placed (Figure 1). All stages mutate a shared `WorldState` container of 2D NumPy arrays (`float32` / `int32`).

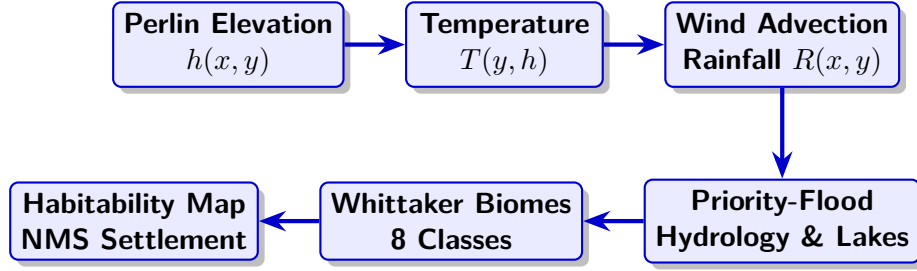


Figure 1: Genesis sequential physical world generation pipeline.

3.1 Terrain Generation

Elevation is generated by summation of multi-octave Perlin noise:

$$h(x, y) = \sum_{k=0}^K A_k \cdot \text{Perlin}\left(\frac{x}{S_k}, \frac{y}{S_k}\right)$$

where amplitude $A_k = 0.5^k$ and scale S_k decreases with octave k . Tiles with $h < h_{\text{sea}} = 0.3$ are classified as ocean.

3.2 Temperature Model

$$T(y, h) = -15 + 47 \cdot \lambda(y) - 38 \cdot \max(0, h) + T_{\text{offset}}$$

where $\lambda(y) = 1 - |y - H/2|/(H/2)$ maps $[0, 1]$ from poles to equator.

3.3 Rainfall – Wind Moisture Advection

For each column x :

$$\text{rain}[x] = m[x] \cdot \left(\frac{0.6}{w} + 35 \cdot \max(0, h_{\text{eff}}[x] - h_{\text{eff}}[x-1]) \right)$$

where $m[x]$ is column air moisture, w is grid width (scaling ensures resolution independence), and $h_{\text{eff}} = \max(h, h_{\text{sea}})$ clamps ocean tiles to sea level. Rain shadows emerge naturally on leeward slopes.

3.4 Biome Classification

Biomes are assigned per cell from a Whittaker temperature–rainfall matrix (Table 1).

Table 1: Whittaker biome classification thresholds

Biome	Temperature (°C)	Rainfall (mm/yr)
Ocean	below sea level	—
Glacier	< -8	any
Tundra	-8 to 0	any
Taiga	0 to 8	≥ 400
Temperate Forest	8 to 20	≥ 700
Grassland	8 to 20	$350\text{--}700$
Desert	any land	< 250
Rainforest	> 20	≥ 1800
Lake	derived from hydrology	—

3.5 Hydrology – Priority-Flood Drainage

Terrain sinks are resolved using the Priority-Flood algorithm [3]: a min-heap initialised from all boundary cells processes interior cells in elevation order, filling depressions to their spillover level. Cells accumulating flow > 1500 units are rivers; filled depressions accumulating > 300 units become lakes.

3.6 Resource and Habitability Layers

Mineral resources (iron, copper) are placed using low-frequency noise masks on high-elevation cells. Five habitability scores (water, food, resource, climate, terrain) are combined as a weighted sum. Circular NMS exclusion (radius 40–45 cells) ensures recommended settlement sites are spatially distributed.

4. Agent Architecture

4.1 Genome

Each agent carries 14 continuous genomic traits, initialised from $\mathcal{U}(0, 1)$ and transmitted with additive Gaussian mutation ($\sigma_{\text{mut}} = 0.05$ per gene per reproduction; Table 2).

4.2 Cognitive Architecture & Deliberation Loop

Agent decisions operate via a homeostatic deliberation system (Figure 2). At each tick, each agent: (1) perceives a radius-bounded neighbourhood and indexes resources into memory; (2) evaluates utility for candidate actions weighted against five drive levels; (3) executes the highest-utility action.

4.3 Drive Dynamics, Mortality, and Reproduction

Five homeostatic drives decay each tick. Mortality occurs when any drive exceeds a terminal threshold. Agents reproduce sexually when hunger and thirst are below thresholds; offspring receive averaged parent genes with additive Gaussian mutation.

Table 2: Genesis 14-gene genomic architecture

Gene	Symbol	Function
Metabolism	<code>g_metabolism</code>	Energy consumption rate
Thermoregulation	<code>g_thermoregulation</code>	Cold/heat exposure resistance
Vision	<code>g_vision</code>	Perception radius
Mobility	<code>g_mobility</code>	Movement speed
Memory Fidelity	<code>g_memory_fidelity</code>	Spatial landmark retention
Planning	<code>g_planning</code>	Utility weighting depth
Novelty Seeking	<code>g_novelty_seeking</code>	Exploration vs. exploitation bias
Social Proximity	<code>g_social_proximity</code>	Colony clustering drive
Aggression	<code>g_aggression</code>	Threat display intensity
Resource Sharing	<code>g_resource_sharing</code>	Altruistic transfer propensity
Risk Sensitivity	<code>g_risk_sensitivity</code>	Danger penalty weight
Resilience	<code>g_resilience</code>	Injury recovery rate
Longevity	<code>g_longevity</code>	Maximum lifespan ceiling
Learning Rate	<code>g_learning_rate</code>	Neural predictor plasticity

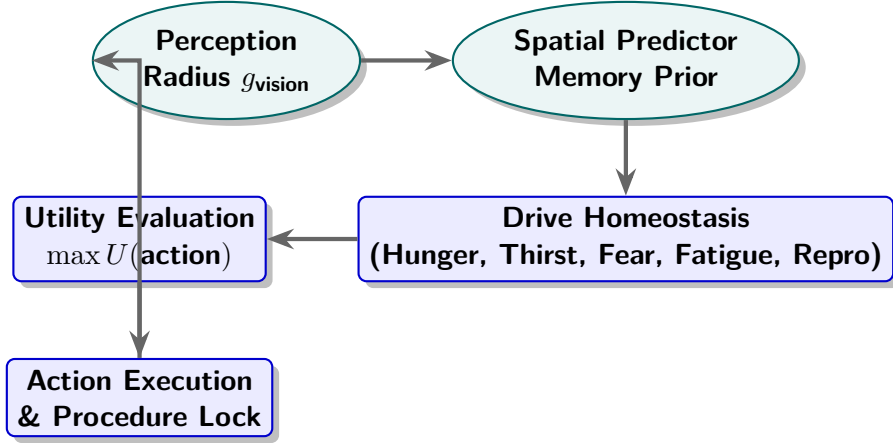


Figure 2: Agent cognitive deliberation and homeostatic drive loop.

4.4 Infrastructure

Agents construct Level-1 shelters (tents) and cache water at reservoir coordinates. Shelters reduce thermal exposure; caches are accessible to agents with high `g_memory_fidelity`.

5. Experimental Protocol

Sixteen founding agents (Gen 0) were divided equally across 4 geographically distinct colonies.

5.1 Experimental Procedure

Seed selection. Seed 1720 was chosen as the canonical pilot seed by running a brief 5,000-tick pre-screen over seeds 1000–2000 and selecting an instance in which all four proposed colony biome placements were genuinely distinct (boreal, arid, temperate, and riparian) and no colony started with water access that would trivially prevent early extinction. The

Table 3: Pilot study experimental configuration

Parameter	Value
Seed	1720
Scarcity multiplier S	3.0 (300% baseline depletion)
Max population $N_{\max}$	200
Mutation rate σ	0.05
Healing multiplier	200×
Disputes	Enabled (resource competition; direct combat disabled)
Climate epoch mode	Default temperate
Grid size	1024 × 1024 cells
World preset	island_chains

seed was fixed before any longitudinal results were observed.

Stopping criterion. The simulation was run until one of two terminal conditions was met: (a) the total living population reached zero, or (b) the tick counter exceeded 5×10^5 ticks. The run terminated at tick 326,075 with 201 agents alive in Colony Beta, triggering a voluntary checkpoint save rather than population collapse.

Telemetry collection. At every tick, the engine logs a compact row to an in-memory buffer that is flushed to Supabase Storage every 1,000 ticks as a compressed CSV chunk. A secondary JSON event stream captures birth, death, reproduction, shelter construction, and inter-agent resource transfer events. Aggregate summary statistics (mean trait values per generation, colony population count, resource totals) are computed online via running accumulators and stored in a separate summary table.

Metrics recorded. Primary outcome metrics are: (i) colony lifespan in ticks and simulated years (1 tick = 0.00278 yr); (ii) maximum generation reached; (iii) total cumulative births over the colony’s lifetime; (iv) per-trait mean and standard deviation at generation 0 and at termination; (v) mortality cause distribution; (vi) infrastructure counts (shelters, water caches). Secondary metrics tracked but not the focus of this pilot include inter-agent dispute frequency, cache utilisation rate, and procedure automation depth.

Hardware-independent execution. All stochastic elements are controlled exclusively through the fixed integer seed passed to a deterministic PRNG (`numpy.random.default_rng(1720)`). The engine contains no wall-clock dependencies, thread-racing, or platform-specific floating-point behaviour; simulation state is therefore exactly reproducible across hardware. Execution speed is not a reproducibility variable.

6. Pilot Study Results

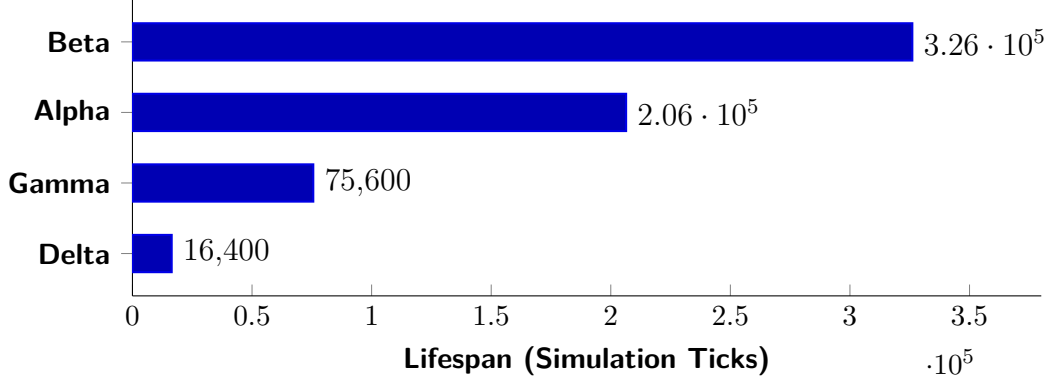
6.1 Colony Setup & Survival Timeline

All colonies began with identical founding genomes (`avg_genetic_diversity` = 0.0 at tick 1; `diversity` = 0.0 implies founding lineages shared identical trait vectors). Total Lifetime Members reports cumulative individual births across the colony’s lifespan, whereas peak concurrent living agents reached 142 in Colony Alpha and 201 in Colony Beta. All 201 final survivors are Beta lineage descendants (founders #4–#7). Figure 3 illustrates colony

Table 4: Colony initial placement and demographic outcome

Colony	Location	Biome	Lifespan	Max Gen	Total Lifetime Members	Status
Delta	(552, 304)	Boreal Highlands	16,400 ticks (45.5 yr)	Gen 1	3	Extinct
Gamma	(847, 540)	Arid Scrubland	75,600 ticks (210 yr)	Gen 8	114	Extinct
Alpha	(146, 503)	Temperate Basin	206,400 ticks (573 yr)	Gen 19	683	Extinct
Beta	(597, 556)	Riparian Oasis	326,075 ticks (905.8 yr)	Gen 38	2,242	Active (201 alive)

survival durations.

Figure 3: Colony lifespans across 326,075 simulation ticks (Seed 1720, $S=3.0$).

7. Results

7.1 Geographic Selection as a Viability Determinant

The four colonies began with identical genome distributions but diverged radically in survival (Table 4).

Colony Delta (Boreal Highlands, tick 16,400): Low ambient temperature ($T_{\text{offset}} < -5^\circ\text{C}$) and sparse food nodes imposed thermal and caloric stress. Only 3 offspring were produced. *Consistent with: boreal biome imposing a viability floor the founding genome did not satisfy.*

Colony Gamma (Arid Scrubland, tick 75,600): Water reserves peaked at 393,944 units (Figure 4). A prolonged drought at Year 180 triggered extinction. *Consistent with: hydrological capacity of the local biome imposing a ceiling behavioural adaptation could not overcome.*

Colony Alpha (Temperate Basin, tick 206,400): Reached a peak of 142 concurrent living agents (683 cumulative lifetime births). Lower mean $g_{\text{vision}} \approx 0.62$ and $g_{\text{memory_fidelity}} \approx 0.51$ (versus Beta’s final 0.79 and 0.61) were associated with failure to access distant food belts. *Single-run design prevents causal attribution.*

Colony Beta (Riparian Oasis, 201 alive): Cumulative births = 2,242. Water storage = 55.5×10^6 units; 3,009 tents and 14 cabins constructed. *Consistent with: hydrological access as the critical viability threshold under $S=3.0$.*

7.2 Directional Genomic Selection & Co-Evolutionary Interpretation

Figure 5 and Table 5 report gene shifts for Colony Beta (Gen 38) relative to pooled founding means ($n=1$; interpret as observation, not estimate).

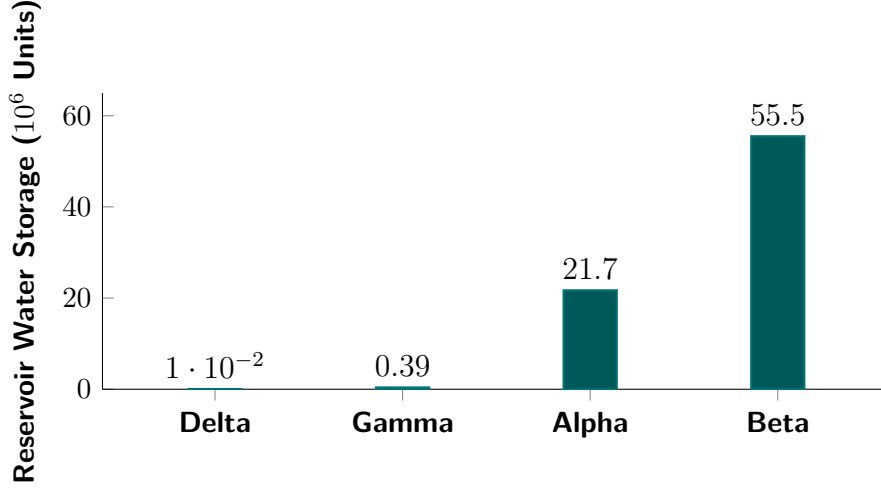


Figure 4: Final accumulated reservoir water storage by colony.

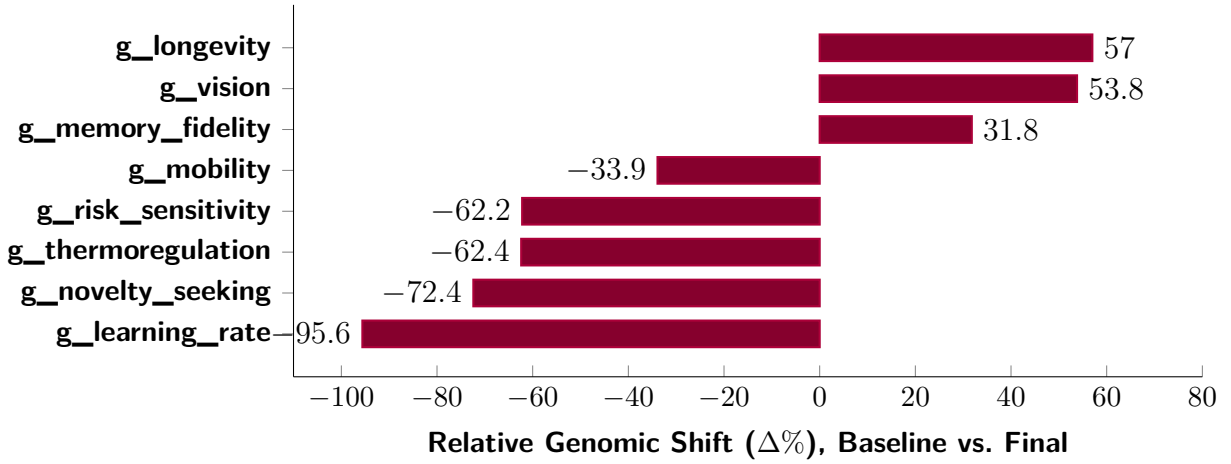


Figure 5: Directional Genomic Shift ($\Delta\%$) across 326,075 ticks (Colony Beta).

Co-Evolutionary Interpretation of Trait Shifts: The joint directional movement across these traits reflects an integrated adaptation strategy:

- **Vision + Longevity** (+53.8%, +57.0%): Long-range perception enabled early detection of distant riparian food patches, while increased longevity allowed experienced agents to maintain shelter networks across decades.
- **Memory Fidelity** (+31.8%): Complemented high vision by anchoring the exact spatial coordinates of Colony Beta’s 55.5×10^6 unit water reservoirs.
- **Mobility + Risk Sensitivity** (−33.9%, −62.2%): As shelter density reached 3,009 tents, high-speed wandering and hyper-vigilance were selected against, favoring energy-conserving, sedentary central-place foraging.

7.3 Cognitive Consolidation: The Learning Rate Collapse

`g_learning_rate` collapsed from 0.520 to 0.023 (−95.6%) alongside 60.96 automated behavioural procedures per agent (maximum observed: 108). This is consistent with the hypothesis that once Colony Beta established a stable spatial network of 3,009 shelters and water reservoirs, high learning rates became counterproductive. Action control literature [2] notes

Table 5: Directional gene shifts — Colony Beta, tick 326,075

Gene	Start	End	$\Delta\%$	Consistent with
<code>g_learning_rate</code>	0.520	0.023	−95.6%	Cognitive consolidation (§7.3)
<code>g_novelty_seeking</code>	0.341	0.094	−72.4%	Exploration penalised under $S=3.0$
<code>g_thermoregulation</code>	0.529	0.199	−62.4%	Technological buffering (§7.4)
<code>g_risk_sensitivity</code>	0.505	0.191	−62.2%	Reduced danger near shelters
<code>g_mobility</code>	0.549	0.363	−33.9%	Sedentary foraging favoured
<code>g_memory_fidelity</code>	0.466	0.614	+31.8%	Water cache retention
<code>g_vision</code>	0.515	0.792	+53.8%	Long-range resource detection
<code>g_longevity</code>	0.540	0.848	+57.0%	Elder knowledge transmission

that habitual action sequences replace goal-directed deliberation in stable environments to minimize cognitive overhead. *Alternative explanations (founder drift, allele fixation) cannot be ruled out from $n=1$.*

7.4 Technological Buffering of Thermoregulation

$g_{\text{thermoregulation}}$ declined −62.4% while direct cold exposure deaths totalled only 6 (0.21% of 2,841). This is consistent with shelter infrastructure creating a warm microenvironment eliminating the survival cost of low thermoregulation—an analogue of cultural buffering [8]. *Confirmation requires an ablation run with shelter construction disabled.*

7.5 Mortality Profile

Of 2,841 deaths: 67.8% old age, 13.6% starvation, 10.5% dehydration, 7.9% injury, 0.2% exposure. Old age dominance (67.8%) indicates multigenerational stability under $S=3.0$.

8. Discussion

8.1 Geography as Independent Variable

Four colonies with identical founding genome distributions produced four qualitatively different survival trajectories, differentiated by initial biome placement. The most parsimonious explanation is geographic. However, small founding populations (4 agents/colony) mean founder sampling variance is substantial.

8.2 The Scarcity–Curiosity Trade-Off

The near-elimination of $g_{\text{novelty_seeking}}$ (−72.4%) under $S=3.0$ is consistent with the hypothesis that exploratory behaviour is inherently expensive under scarcity. A cross-scarcity design ($S \in \{1.0, 2.0, 3.0, 5.0\}$) would test whether this decline is monotonically related to scarcity intensity.

8.3 Cognitive Consolidation and Procedural Fixation

The learning rate collapse alongside high procedure count suggests an evolutionary analogue of skill consolidation: a transition from deliberate cognitive exploration to habituated

procedural execution [2]. Whether this parallel to mammalian habit formation is meaningful or superficial is an open question.

9. Open Questions

1. Does Beta always outcompete Alpha/Gamma on seed 1720? (10× re-runs.)
2. Is $g_{\text{novelty_seeking}}$ collapse monotonic with S ? ($S \in \{1.0, 2.0, 3.0, 5.0\}$.)
3. Does disabling shelter construction eliminate thermoregulation decay?
4. Would direct combat prevent Beta’s single-clade monopolisation?
5. Do low-curiosity, low-thermoregulation lineages survive sudden glacial epoch transitions?

10. Code, Data, and Replay Availability

To ensure complete scientific transparency and independent verification, all components of Project Genesis are open access:

Source Code: The simulation engine, world generator, and agent modules are available at <https://github.com/vinaykhosya/genesis-civilizations>.

Research Portal & Archive: Interactive visual exploration, metadata, experiment archives, and JSON-LD schemas are hosted at <https://genesis.vinaykhosya.com/archive>.

Experiment Telemetry: Full tick-by-tick CSV logs, summary statistics, and event streams for seed 1720 are archived in Supabase Storage with signed URLs accessible via the research portal.

Replay Viewer: Client-side Canvas2D replay streams enabling tick-level spatial inspection of agent movements, shelter construction, and colony boundaries are accessible directly through the civilization detail view at <https://genesis.vinaykhosya.com/archive>.

License: Code is released under the MIT License; experimental data and paper documents are published under the Creative Commons Attribution 4.0 International (CC BY 4.0) License.

11. Limitations

1. $n=1$. Single run, single seed. No result should be interpreted as general.
2. **Founder sampling variance.** 4 agents per colony; colony performance differences cannot be cleanly attributed to geography.
3. **No statistical testing.** Multi-run data will enable Mann-Whitney U and Cohen’s d analysis.
4. **Selection vs. drift.** Directional selection cannot be distinguished from drift without replicate runs.

5. **Simulation fidelity.** Temperature uses a smooth latitude factor; cognition modelled as utility functions, not neural networks.
6. **Parameter sensitivity.** Mutation rate, scarcity, and population ceiling were fixed throughout.

12. Conclusion

Genesis is a physically grounded artificial life simulator in which terrain, climate, hydrology, and biome distribution are derived from physical models rather than set by the researcher. The pilot study on seed 1720 ($S=3.0$, 326,075 ticks) provides preliminary evidence that geographic placement is a primary determinant of colony viability when founding genomes are identical, and that directional selection under scarcity produces characteristic cognitive signatures—particularly the suppression of exploratory behaviour and the consolidation of procedural routines at the cost of neural plasticity.

Genesis is not presented as a model of biological reality, but as an experimental framework for studying how geography, cognition, and evolution interact over long temporal horizons under reproducible computational conditions. Full simulation code, experiment data, and a public research archive are available at <https://genesis.vinaykhosya.com>.

The current work establishes Genesis as a reproducible experimental platform. Immediate next steps include multi-seed replication ($n \geq 10$, seeds drawn from a pre-registered list) to distinguish geographic effects from founder sampling variance; systematic parameter ablations varying scarcity ($S \in \{1.0, 2.0, 3.0, 5.0\}$) and mutation rate; a shelter-disabled ablation to test the technological buffering hypothesis; and glacial epoch transition experiments to probe the long-term stability of low-plasticity lineages. Longer-term work will compare Genesis evolutionary outcomes against reinforcement-learning baselines and heuristic foraging agents operating in identical physical environments, and will target a statistically rigorous follow-up study powered by sufficient replication to support Mann–Whitney U tests and Cohen’s d effect-size estimation across all primary outcome metrics.

References

- [1] Baker, B., Kanitscheider, I., Markov, T., Zheng, Y., Zhou, G., Bowen, C., & Mordatch, I. (2020). Emergent tool use from multi-agent autocurricula. *International Conference on Learning Representations (ICLR)*.
- [2] Balleine, B. W., & O’Doherty, J. P. (2010). Human and rodent homologs in action control: Corticostriatal determinants of goal-directed and habitual action. *Neuropsychopharmacology*, 35(1), 48–69.
- [3] Barnes, R., Lehman, C., & Mulla, D. (2014). Priority-flood: An optimal depression-filling and watershed-labeling algorithm for digital elevation models. *Computers & Geosciences*, 62, 117–127.
- [4] Berner, C., et al. (2019). Dota 2 with large scale deep reinforcement learning. *arXiv:1912.06680*.
- [5] Channon, A. (2001). Passing the ALife test: Activity statistics classify evolution in Geb as unbounded. *Advances in Artificial Life*, 417–426.

- [6] Ecoffet, A., Huizinga, J., Lehman, J., Stanley, K. O., & Clune, J. (2021). First explore, then exploit: Go-Explore solves hard exploration problems. *Nature*, 590, 580–586.
- [7] Epstein, J. M., & Axtell, R. (1996). *Growing Artificial Societies: Social Science from the Bottom Up*. MIT Press.
- [8] Laland, K. N., Odling-Smee, J., & Feldman, M. W. (2000). Niche construction, biological evolution, and cultural change. *Behavioral and Brain Sciences*, 23(1), 131–146.
- [9] Lehman, J., & Stanley, K. O. (2011). Abandoning objectives: Evolution through the search for novelty alone. *Evolutionary Computation*, 19(2), 189–223.
- [10] Lenski, R. E., Ofria, C., Pennock, R. T., & Pennock, D. (2003). The evolutionary origin of complex features. *Nature*, 423, 139–144.
- [11] Ofria, C., & Wilke, C. O. (2004). Avida: A software platform for research in computational evolutionary biology. *Artificial Life*, 10(2), 191–229.
- [12] Ray, T. S. (1991). An approach to the synthesis of life. *Artificial Life II*, 371–408.
- [13] Stanley, K. O., & Miikkulainen, R. (2002). Evolving neural networks through augmenting topologies. *Evolutionary Computation*, 10(2), 99–127.
- [14] Vinyals, O., et al. (2019). Grandmaster level in StarCraft II using multi-agent reinforcement learning. *Nature*, 575, 350–354.
- [15] Whittaker, R. H. (1975). *Communities and Ecosystems* (2nd ed.). Macmillan.
- [16] Yaeger, L. (1994). Computational genetics, physiology, and artificial life in Polyworld. *Artificial Life III*, 263–298.